

- 3 (a) On one particular ride in an amusement park, passengers in a carriage car 'loop-the-loop' in a vertical circle, as illustrated in Fig. 3.1.

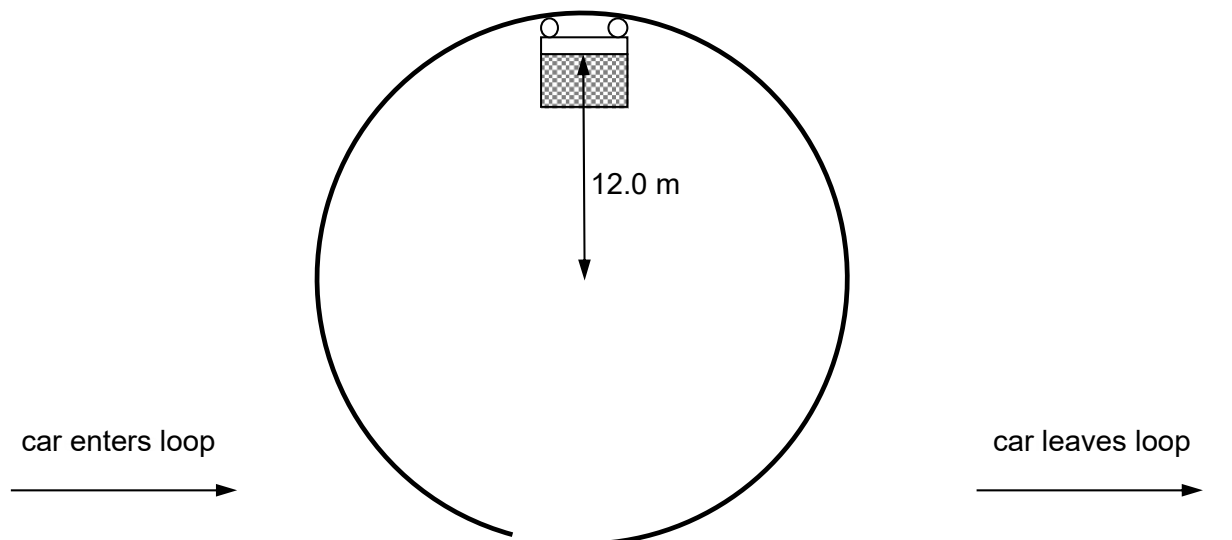


Fig. 3.1

The car moves freely, so there is no frictional forces and no driving force propelling the car when it is moving in the loop.

The centre of gravity of the car moves in a circle with radius 12.0 m.

Determine

- (i) the minimum speed of the carriage car at the top of the circular track shown in Fig. 3.1,

minimum speed = m s^{-1} [3]

- (ii) the minimum speed of the car leaving the loop.

minimum speed = m s^{-1} [2]

- (b) In another ride in the same amusement park, passengers sit in carriages which travel at variable angular velocities in a vertical circle. Fig. 3.2 shows the motion of one of the carriages.

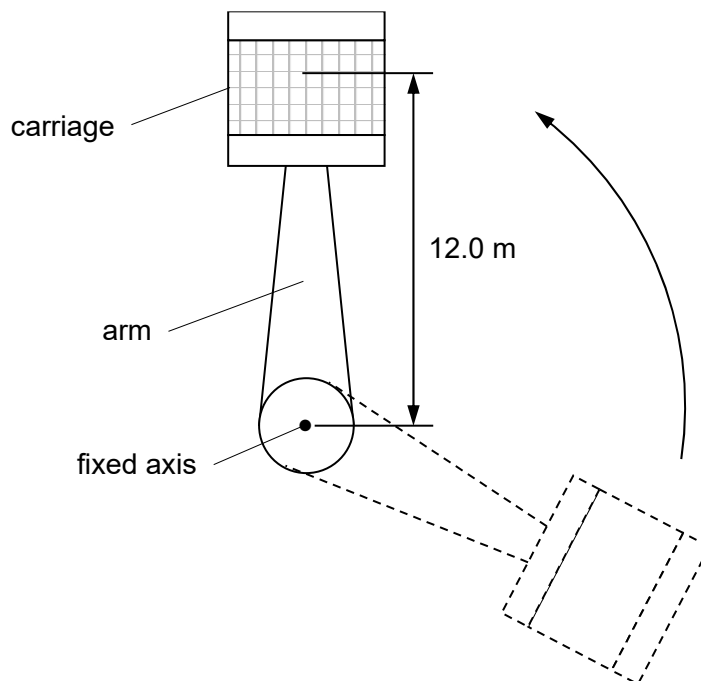


Fig. 3.2

The centre of gravity of the carriage moves in a circle with radius 12.0 m.

- (i) State what is meant by the *angular velocity*.

.....
.....
.....[2]

- (ii) Suggest why it is possible for the carriage in Fig. 3.2 to be at rest (i.e. zero angular velocity) at the top of the circle whereas it is impossible for the car in (a).

.....
.....
.....
.....[2]